HAT-P-36b

R-0958

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_HATP-36_230212` · created 2026-07-29T15:29:47+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459987.8927 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1548 +/- 0.0061
Transit depth (Rp/Rs)^2	2.4 +/- 0.19 [%]
Orbital Inclination (inc)	81.95 +/- 1.68
Airmass coefficient 1 (a1)	1.076 +/- 0.01
Airmass coefficient 2 (a2)	-0.07 +/- 0.023
Scatter in the residuals of the lightcurve fit is	0.94 %
Best Comparison Star	None
Optimal Aperture	4.1
Optimal Annulus	6.89
Transit Duration (day)	0.0843 +/- 0.0073

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1548 ± 0.0061 vs 0.126 (deviation 4.64σ)
Duration fit vs published	0.0843 d vs 0.0925 d (deviation 1.1σ)
Mid-time O–C	-12.7 min
Depth SNR	25.0
Residual scatter	0.94 %

Light curve

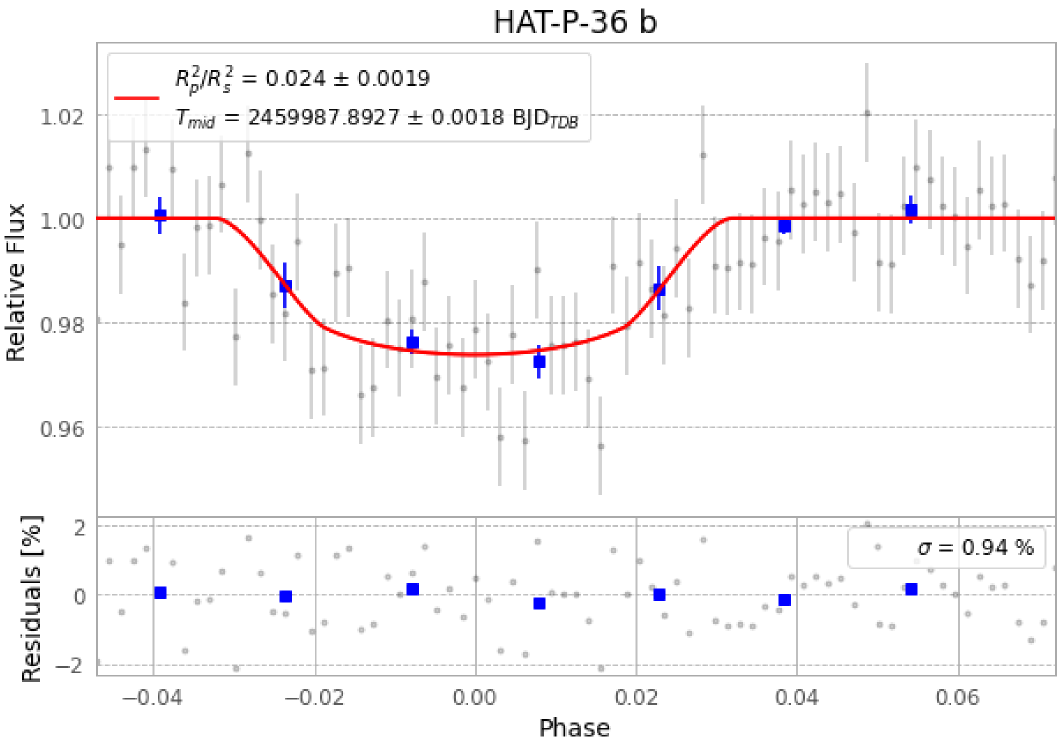

Figure 1 — FinalLightCurve_HAT-P-36 b_2023-02-12.png (SHA-256 dd804f8e6065e03b…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [524, 132], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 65bd95f27029c181…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

Data provenance

77 FITS frames (51 MB), `HATP-36230212074815.FITS` → `HATP-36230212113615.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-230212.log (7669d7db3bc1…), FinalLightCurve_HAT-P-36 b_2023-02-12.png (dd804f8e6065…), FinalLightCurve_HAT-P-36 b_2023-02-12.csv (f25393d127c5…)

Compute resources

Wall time 145.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

